ELEC70070
EE9-SO20

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc and EEE/EIE PART IV: MEng and ACGI

INFORMATION THEORY

Friday, 10 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : C. Ling
Second Marker(s) : J Barrett

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Information for students

Notation:

(a) Random variables are shown in Tahoma font. x, $\mathbf{x}$, $\mathbf{X}$ denote a random scalar, vector and matrix respectively.

(b) The size of a set A is denoted by $|A|$.

(c) By default, the logarithm is to the base 2.

(d) $\oplus$ denotes the exclusive-or operation, or modulo-2 addition.

(e) "i.i.d." means "independent identically distributed".

(f) $H(\cdot)$ is the entropy function.

(g) $C(x) = \frac{1}{2}\log_2(1+x)$ is the capacity function for the Gaussian channel in bits/channel use.

The Questions

1. Basics of information theory.

a) Let the joint distribution of two random variables x and y be given by

$p(x,y)$	$y=0$	$y=1$
$x=0$	1/3	1/3
$x=1$	0	1/3

Compute:

i) The entropy $H(x)$, $H(y)$

ii) The conditional entropy $H(x|y)$, $H(y|x)$

iii) The joint entropy $H(x,y)$

iv) The mutual information $I(x,y)$

v) Draw a Venn diagram for the above quantities.

[15]

b) A fair coin is flipped until the first head occurs. Let x denote the number of flips required. The following equalities may be useful.

$$\sum_{n=1}^{\infty} r^n = \frac{r}{1-r} \qquad \sum_{n=1}^{\infty} n r^n = \frac{r}{(1-r)^2}.$$

i) Find the entropy $H(x)$ in bits.

ii) A random variable x is drawn according to this distribution. To determine x, we ask a sequence of yes-no questions of the form "Is $x = i$?" for $i = 1, 2, 3, \ldots$. Find the expected number of questions required to determine x.

[10]

2. Source coding/Fano inequality.

a) Fano inequality. Consider the Markov chain shown in Fig. 2.1, where x and y are discrete random variables, and $\hat{x}$ is the estimate of x.

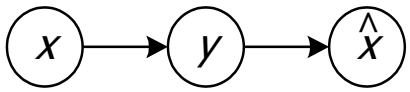


Fig. 2.1. Markov chain arising in Fano's inequality.

Define a random variable $e = (\hat{x} \neq x) \in \{0,1\}$. Justify each step of the following derivations.

$$
\begin{aligned}
H(e,x \mid y) &\overset{(1)}{=} H(x \mid y) + H(e \mid x,y) \overset{(2)}{=} H(e \mid y) + H(x \mid e,y) \\
&\overset{(3)}{\Rightarrow} H(x \mid y) + 0 \leq H(e) + H(x \mid e,y) \\
&\overset{(4)}{=} H(e) + H(x \mid y, e=0)(1-p_e) + H(x \mid y, e=1)\, p_e \\
&\overset{(5)}{\leq} H(p_e) + 0 \times (1-p_e) + \log(|\mathsf{X}|-1)\, p_e \\
&\overset{(6)}{\Rightarrow} p_e \geq \frac{\left(H(x \mid y) - H(p_e)\right)}{\log\left(|\mathsf{X}|-1\right)} \overset{(7)}{\geq} \frac{\left(H(x \mid y) - 1\right)}{\log\left(|\mathsf{X}|-1\right)}
\end{aligned}
$$

[10]

b) Polar codes for source coding. Let the source bits be i.i.d. Bernoulli(p), where p = 0.11. Consider the following polar transform:

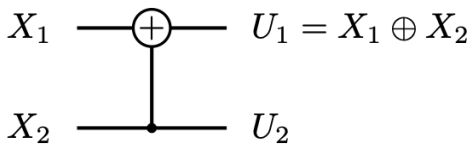


Fig. 2.2. Basic polar transform.

i) Compute $H(U_1), H(U_2|U_1)$ using conservation of entropy. [6]

ii) Find the conditional distribution $P(U_2|U_1)$ and use it to recalculate $H(U_2|U_1)$. [9]

3. Channel capacity.

a) Consider the Gaussian channel with feedback shown in Fig. 3.1.

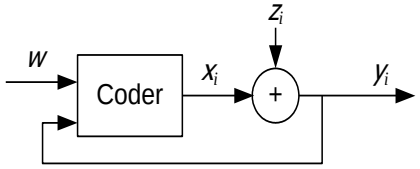


Fig. 3.1. Gaussian channel with feedback.

Justify each step of the following proof that feedback increases the capacity by ½ bits at most.

$$I(\mathrm{w};\mathrm{y}) = h(\mathrm{y}) - h(\mathrm{y}|\mathrm{w}) = h(\mathrm{y}) - \textstyle\sum_{i=1}^{n} h(\mathrm{y}_i|\mathrm{w},\mathrm{y}_{1:i-1})$$

$$\overset{(1)}{=} h(\mathrm{y}) - \sum_{i=1}^{n} h(\mathrm{y}_i|\mathrm{w},\mathrm{y}_{1:i-1},\mathrm{x}_{1:i},\mathrm{z}_{1:i-1})$$

$$\overset{(2)}{=} h(\mathrm{y}) - \sum_{i=1}^{n} h(\mathrm{z}_i|\mathrm{w},\mathrm{y}_{1:i-1},\mathrm{x}_{1:i},\mathrm{z}_{1:i-1})$$

$$\overset{(3)}{=} h(\mathrm{y}) - \sum_{i=1}^{n} h(\mathrm{z}_i|\mathrm{z}_{1:i-1}) = h(\mathrm{y}) - h(\mathrm{z}) \overset{(4)}{\leq} \frac{1}{2}\log_2\frac{|\mathbf{K}_\mathrm{y}|}{|\mathbf{K}_\mathrm{z}|}$$

$$C_{n,FB} \overset{(5)}{=} \max_{\mathrm{tr}(\mathbf{K}_\mathbf{x})\leq nP} \tfrac{1}{2} n^{-1}\log_2\frac{|\mathbf{K}_\mathbf{y}|}{|\mathbf{K}_\mathbf{z}|} \leq \max_{\mathrm{tr}(\mathbf{K}_\mathbf{x})\leq nP} \tfrac{1}{2} n^{-1}\log_2\frac{|2(\mathbf{K}_\mathbf{x}+\mathbf{K}_\mathbf{z})|}{|\mathbf{K}_\mathbf{z}|}$$

$$\overset{(6)}{=} \max_{\mathrm{tr}(\mathbf{K}_\mathbf{x})\leq nP} \tfrac{1}{2} n^{-1}\log_2\frac{2^n|\mathbf{K}_\mathbf{x}+\mathbf{K}_\mathbf{z}|}{|\mathbf{K}_\mathbf{z}|} \overset{(7)}{=} \tfrac{1}{2} + \max_{\mathrm{tr}(\mathbf{K}_\mathbf{x})\leq nP} \tfrac{1}{2} n^{-1}\log_2\frac{|\mathbf{K}_\mathbf{x}+\mathbf{K}_\mathbf{z}|}{|\mathbf{K}_\mathbf{z}|} \overset{(8)}{=} \tfrac{1}{2} + C_n$$

[10]

b) A channel with alphabet {0, 1, 2, 3, 4} has transitional probabilities of the form

$$p(y|x) = \begin{cases} 1/2 & \text{if } y = x \pm 1 \mod 5 \\ 0 & \text{otherwise} \end{cases}$$

where mod 5 denotes the modulo 5 operation. Compute the channel capacity of this channel.

[5]

c) Find the capacity of the following discrete memoryless channel with probability transition matrix

$$P_{Y|X} = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & \frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

[10]

4. Network information theory.

a) Multi-access channel.

i) With the help of a diagram, explain the term multi-access channel. Give an example.

ii) Write down the capacity region of the two-user bandlimited multi-access channel with additive white Gaussian noise, where the users have equal powers P, bandwidth W, and the noise power is N. Draw the region and explain how to achieve the two corner points.

iii) Find the capacity region for this two-user multiple access channel with infinite bandwidth W. Show that both senders can send at their individual capacities, i.e., infinite bandwidth eliminates interference.

[15]

b) Slepian-Wolf region for binary sources. Let X be Bernoulli(p). Let Z be Bernoulli(r), and let Z be independent of X. Finally, let $Y = X \oplus Z$ (mod 2 addition). Let X be described at rate R_1 and Y be described at rate R_2. Calculate and sketch the Slepian-Wolf rate region for this source pair (X, Y).

[10]